Supplementary material to 'The Devil's Dung? Money as a mechanism of generalized reciprocity in human societies'

Section 1 provides the complete model pseudocode, while Section 2 presents the flowchart. Section 3 includes sensitivity analysis and robustness tests.

1 Model Pseudocode

```
In an agent-based environment,
1. Setup:
        1.1 Define global parameters:
                set BENEFIT-TO-COST-RATIO [variable input > 1]
                set COST 1
                determine BENEFIT [BENEFIT-TO-COST-RATIO * COST]
                set INITIAL-STANDING [variable (binary) input, "good" by default]
                set INITIAL-LIQUIDITY [variable input > 0]
                set COOPERATION-COUNTER 0
        1.2 Create agent population:
                1.2.1 Define agent types, create a given number of agents for each type:
                        Create Unconditional Cooperators [variable input, 100 by default]
                        Create Defectors [variable input, 100 by default]
                        Create Direct-Reciprocators [variable input, 100 by default]
                        Create Indirect-Reciprocators [variable input, 100 by default]
                        Create Money-Users [variable input, 100 by default]
                1.2.2 Define initial agent variables. For every agent:
                        set ID [unique identifier]
                        set fitness 0
                        set current-partner NA [ID of another agent]
                        set memory [empty list]
                        set standing [INITIAL-STANDING, "good" by default]
                        set money-balance [result of assign-initial-money function]
        1.3 Define auxiliary functions:
                function help-partner():
                        decrease acting-agent's fitness by COST
                        increase current-partner's fitness by BENEFIT
                        set acting-agent's standing to "good"
                        increase COOPERATION-COUNTER by 1
                function do-not-help():
                        add acting-agent to current-partner's memory
                         if current-partner's standing is "good":
                           set acting-agent's standing to "bad"
                function assign-initial-money():
                        if INITIAL-LIQUIDITY < 1:
                                if random uniform (0,1) < INITIAL-LIQUIDITY:
                                        set money-balance 1
                                else:
                                        set money-balance 0
                        else:
                                set money-balance INITIAL-LIQUIDITY
2. Dynamics: at every time step, iterate through all agents twice:
        2.1 First iteration: for each agent,
                set fitness 0
                set current-partner = a randomly selected other agent
        2.2 Second iteration: for each agent,
                if type = "Unconditional Cooperator":
                        help_partner()
                if type = "Defector":
                        do_not_help()
                if type = "Direct-Reciprocator":
                        if current-partner is not in memory:
                                help_partner()
```

```
                else:
                        do_not_help()
                        remove current-partner from memory
        if type = "Indirect-Reciprocator":
                if standing of current-partner is "good":
                        help_partner()
                else:
                        do_not_help()
        if type = "Money-User":
                if money-balance of current-partner > 0:
                        help_partner()
                        increase acting-agent's money-balance by 1
                        decrease current-partner's money-balance by 1
                else:
                        do_not_help()
3. Strategy evolutionary updating: After all agents have interacted, a random agent is
selected to potentially change its strategy. This agent may adopt a new strategy based on
the success of other surviving strategies in the current round, as in a roulette wheel
selection process. The probability of adopting a specific strategy is weighted according to
its relative in-round fitness. To avoid negative probabilities, each agent's fitness is
increased by one unit at the end of each time step.
4. Collect metrics: At the end of each iteration, collect data on the number of agents
following each strategy and the total cooperation rate observed in the population. After
data collection, reset COOPERATION-COUNTER to 0.
5. Record metrics: Every 250 iterations, log the number of agents following each strategy
and the overall cooperation rate in the population to a .csv file.
The simulation is interrupted after a predetermined number of time steps (30000 by default).
```

2 Model Flowchart

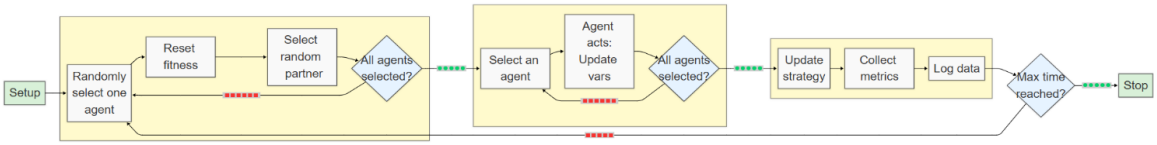

Figure S1. Model execution flowchart. Red squares indicate "No", while green circles indicate "Yes" decisions at branching points.

3 Sensitivity Analysis

Figure S2 shows the results of the evolutionary tournament in the benchmark scenario, where agents following the monetary exchange strategy were removed from the initial population. The population size was fixed at $N = 500$, with an equal distribution of strategies: 125 unconditional cooperators, 125 direct reciprocators, 125 indirect reciprocators, and 125 defectors.

Given the settings of our tournament, defectors are consistently eliminated from the population, even when the returns to cooperation are low. This occurs because standing-based indirect reciprocators refrain from cooperating with them: defectors acquire a "bad" standing, and refusing to help such individuals constitutes a justified defection that does not damage the reciprocators' reputation. As a result, indirect reciprocators can continue to cooperate with each other within the reputation circuit (see also the Results section in the main manuscript), while defectors remain excluded from it. Nevertheless, equilibrium populations very often consist of a

mix of unconditional cooperators and indirect reciprocators, with the relative share of unconditional cooperators increasing as the returns to cooperation become very high.

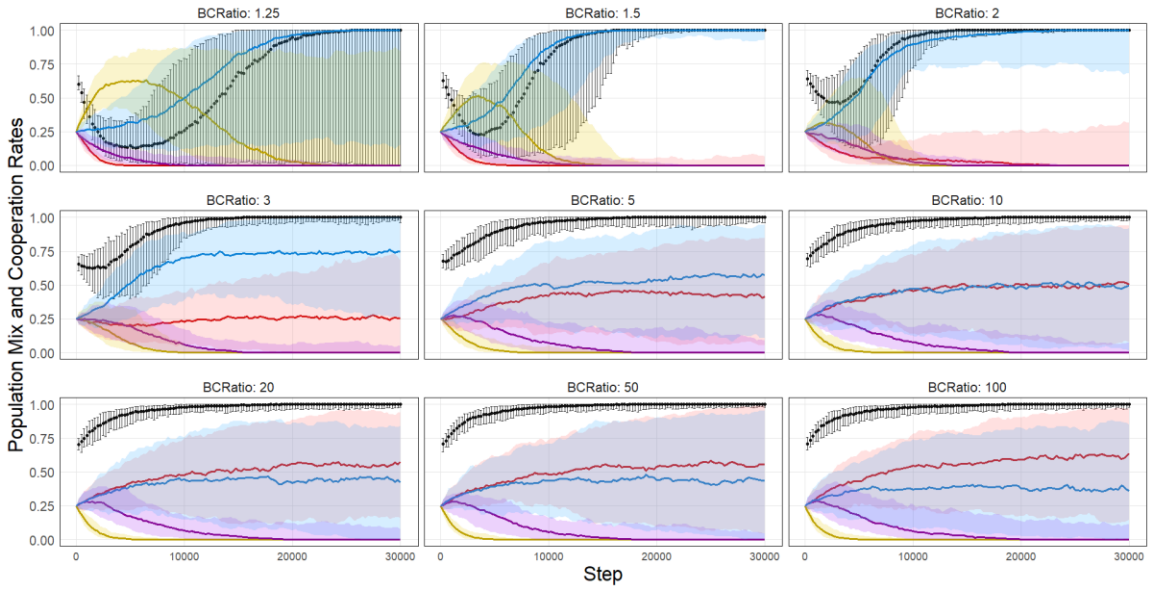

Figure S2. Evolutionary dynamics of population composition and cooperation in the benchmark tournament model under selected benefit-to-cost ratios. In this scenario, agents using the monetary exchange strategy were removed from the initial population, and the prevalence of the remaining four strategies was scaled proportionally. Colored lines represent the median population share of each surviving strategy over time, with shaded ribbon bands indicating interquartile ranges (IQRs). Specifically, yellow denotes defectors, red unconditional cooperators, light blue indirect reciprocators, and purple direct reciprocators. The black line and associated error bars show the median and IQR of the cooperation rate in the population. Each curve is based on 100 independent simulation runs per condition, each lasting 30000 iterations, with data recorded every 250 steps.

For this reason, unconditional cooperators can be seen as *neutral mutants* of the standing-based indirect reciprocity strategy. Since indirect reciprocators cannot eliminate them from the population, these unconditional cooperators persist and ultimately render the system vulnerable to late-stage invasions by defectors or by mutations that favor defection. In contrast, the monetary exchange strategy effectively excludes unconditional cooperators (see Figure 5 in the main manuscript), thereby enhancing the system's resilience to subsequent invasions by defectors.

Figure S3 illustrates these dynamics in detail. For simplicity, and without loss of generality, we assume the same benefit-to-cost ratios and liquidity levels as in Figure 2 of the main manuscript. The tournament setup is identical to that in Figure 2, with the exception that, at step 15000, half of the agents employing the monetary exchange strategy are randomly converted into defectors.

By the time this mass invasion occurs, the monetary exchange strategy has already become an evolutionarily stable under all parameter combinations shown in Figure S3, and cooperation rates have stabilized at levels that vary with initial liquidity (see also Figures 2 and 3 of the main manuscript). Since agents are randomly converted, there is an equal probability of selecting individuals with high or low balances. This results in a variable number of defectors with high balances, which, as discussed in the main manuscript, poses a threat to the stability of money-mediated cooperative circuits.

Nevertheless, following this shock, defectors are once again negatively selected in favor of the monetary exchange strategy. Some transient instability emerges when the benefits to cooperation are very low and liquidity is high, yet, on average, defectors are ultimately eliminated. Cooperation rates return to their pre-invasion equilibrium levels at varying speeds, underscoring the robustness of the monetary exchange strategy to late-stage invasions by defectors.

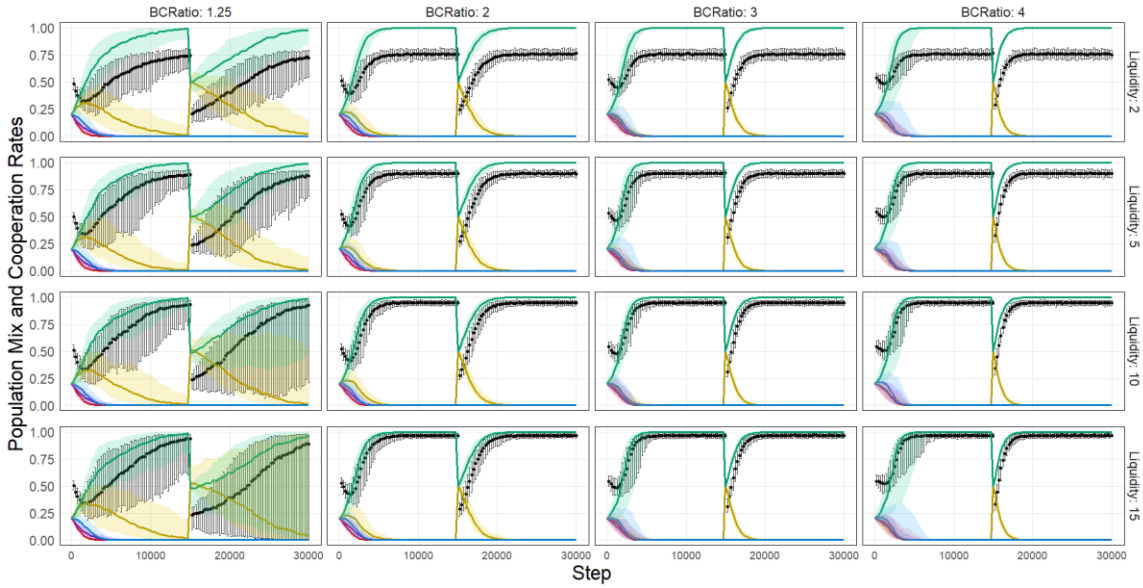

Figure S3. Robustness of the monetary exchange strategy to a mass invasion of defectors. This figure replicates the tournament setup from Figure 2 of the main manuscript, with one important difference: at the 15000th simulation step, half of the agents following the monetary exchange strategy were forcibly converted into defectors at random. Colored lines represent the median population share of each surviving strategy over time, with shaded ribbon bands indicating interquartile ranges (IQRs). Specifically, green indicates agents following the monetary exchange strategy, yellow indicates defectors, red indicates unconditional cooperators, light blue indicates indirect reciprocators, and purple indicates direct reciprocators. The black line and associated error bands indicate the median and IQR of the cooperation rate in the population. Each curve is based on 100 independent simulation runs per condition, each lasting 30000 iterations, with data recorded every 250 steps.

Figures S4 and S5 provide additional tests of the robustness of the monetary exchange strategy by examining its stability when forcibly converting half of the agents into either unconditional cooperators (Figure S4) or a mix of unconditional cooperators and defectors (Figure S5). Consistent with the results in Figure 5 of the main manuscript, Figure S4 shows that money is not a neutral mutant of unconditional cooperators, as they are rapidly eliminated from the population after the shock.

Figure S5 shows that converting half of the agents following the monetary exchange strategy into a mix of defectors and unconditional cooperators has no lasting effect on the stability of the system. In this scenario, the forced mutation introduces defectors who benefit from the unconditional help of cooperators, making defection temporarily even more fit than in the case shown in Figure S3. However, this advantage is not sufficient to destabilize the money-induced cooperation circuit, which remains resilient to the shock.

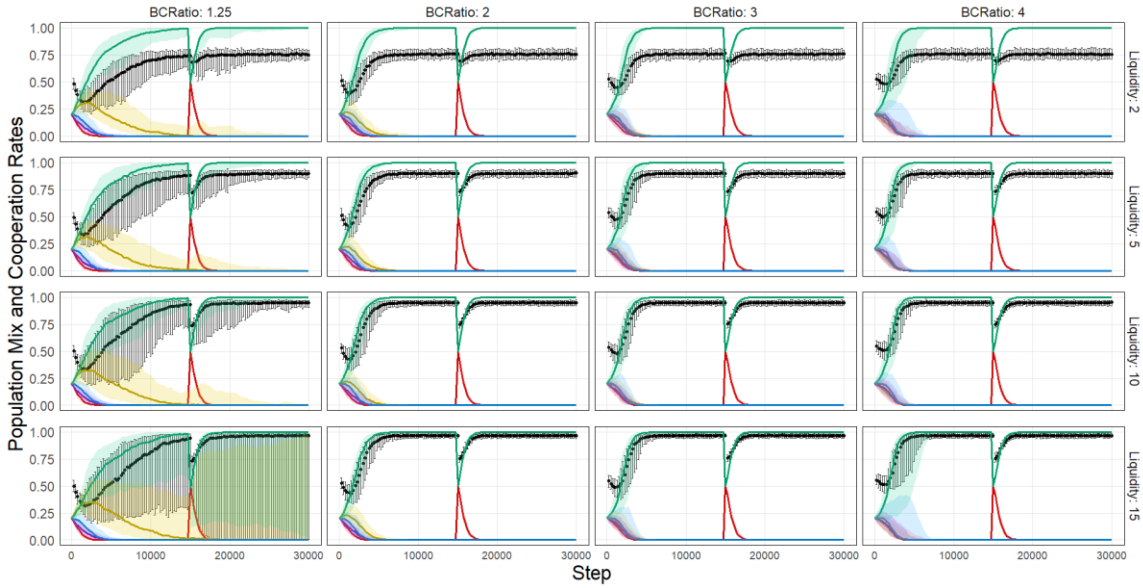

Figure S4. Robustness of the monetary exchange strategy to a mass invasion of unconditional cooperators. This figure replicates the tournament setup from Figure 2 of the main manuscript, with one key difference: at the 15000th simulation step, half of the agents following the monetary exchange strategy were forcibly converted into unconditional cooperators at random. Colored lines represent the median population share of each surviving strategy over time, with shaded ribbon bands indicating interquartile ranges (IQRs). Specifically, green indicates agents following the monetary exchange strategy, yellow indicates defectors, red indicates unconditional cooperators, light blue indicates indirect reciprocators, and purple indicates direct reciprocators. The black line and associated error bands indicate the median and IQR of the cooperation rate in the population. Each curve is based on 100 independent simulation runs per condition, each lasting 30000 iterations, with data recorded every 250 steps.

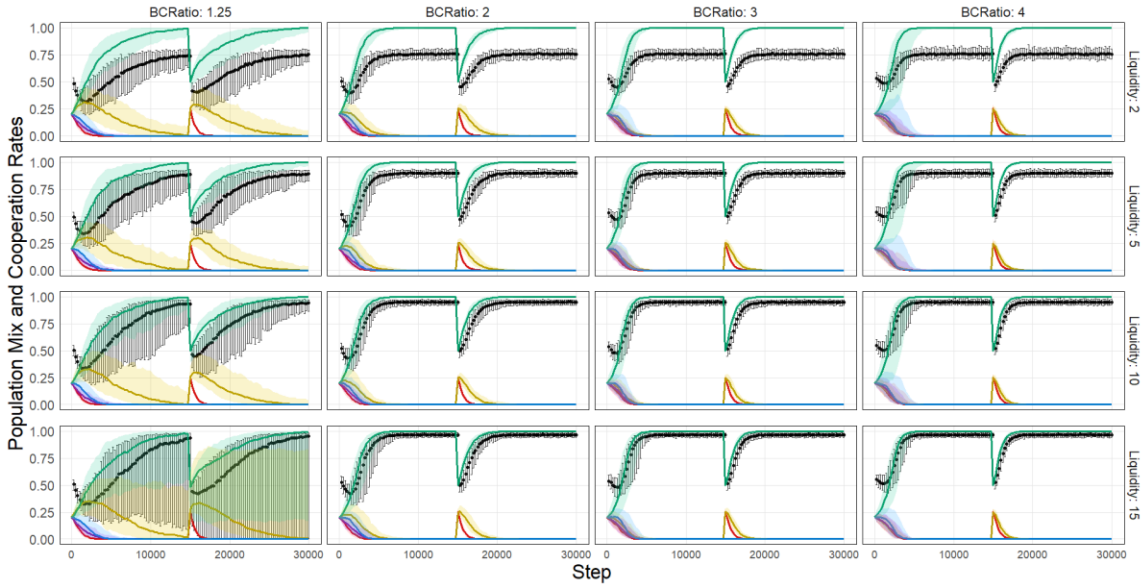

Figure S5. Robustness of the monetary exchange strategy to a mass invasion of a mix of unconditional cooperators and defectors. This figure replicates the tournament setup from Figure 2 of the main manuscript, with one key difference: at the 15000th simulation step, half of the agents following the monetary exchange strategy were forcibly converted into either unconditional cooperators or defectors at random. Colored lines represent the median population share of each surviving strategy over time, with shaded ribbon bands indicating interquartile ranges (IQRs). Specifically, green indicates agents following the monetary exchange strategy, yellow indicates defectors, red indicates unconditional cooperators, light blue indicates indirect reciprocators, and purple indicates direct reciprocators. The black line and associated error bands indicate the median and IQR of the cooperation rate in the population. Each curve is based on 100 independent simulation runs per condition, each lasting 30000 iterations, with data recorded every 250 steps.

Figures S6 and S7 illustrate the evolutionary trajectories and cooperation rates in populations initially composed exclusively of unconditional cooperators, defectors, and agents adopting the monetary exchange strategy. In Figure S6, the simulation starts with 167 agents per strategy (based on rounding up $N = 500$ divided by three), while in Figure S7 the initial population is increased by 20% to $N = 600$, with 200 agents per strategy. Across all scenarios examined, the monetary exchange strategy consistently outperforms the other two, ultimately emerging as the evolutionarily stable strategy. This dominance is driven by the formation of money-mediated cooperation circuits that systematically exclude both defectors and unconditional cooperators from participating in cooperative exchanges.

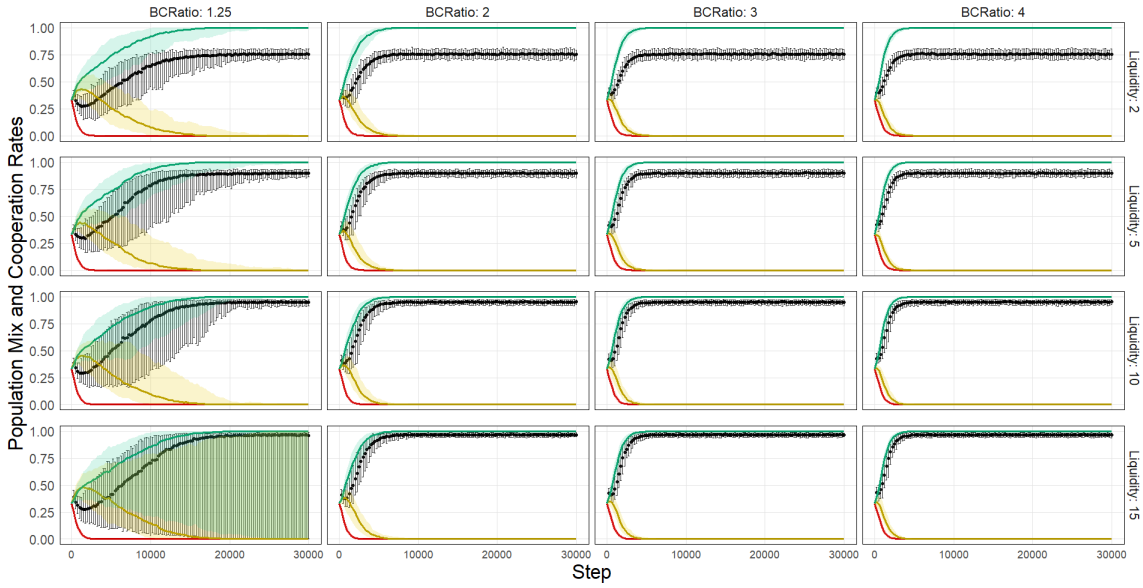

Figure S6. Evolutionary dynamics of population composition and cooperation in the tournament model under selected benefit-to-cost ratios. In this scenario, the initial population consists exclusively of unconditional cooperators (red), defectors (yellow), and agents following the monetary exchange strategy (green), with reciprocity strategies removed. The simulation starts with 167 agents per strategy. Colored lines represent the median population share of each surviving strategy over time, with shaded ribbon bands indicating interquartile ranges (IQR). The black line and associated error bars show the median and IQR of the cooperation rate in the population. Each curve is based on 100 independent simulation runs per condition, each lasting 30000 iterations, with data recorded every 250 steps.

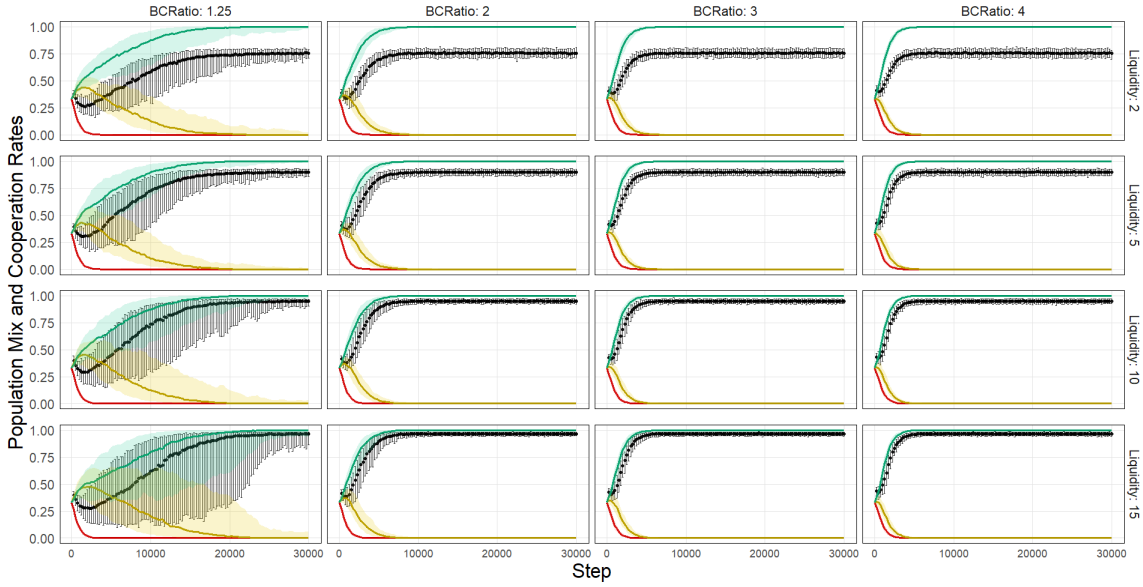

Figure S7. Evolutionary dynamics of population composition and cooperation in the tournament model under selected benefit-to-cost ratios. In this scenario, the initial population consists exclusively of unconditional cooperators (red), defectors (yellow), and agents following the monetary exchange strategy (green), with reciprocity strategies removed. The simulation starts with 200 agents per strategy. Colored lines represent the median population share of each surviving strategy over time, with shaded ribbon bands indicating interquartile ranges (IQR). The black line and associated error

bars show the median and IQR of the cooperation rate in the population. Each curve is based on 100 independent simulation runs per condition, each lasting 30000 iterations, with data recorded every 250 steps.

Figures S8 and S9 replicate Figure 4 of the main manuscript, showing the equilibrium cooperation rates and the dominant strategy across all 140 combinations of liquidity and benefit-to-cost ratios. These simulations assume initial populations consisting of either 167 (Figure S8) or 200 (Figure S9) unconditional cooperators, defectors, and agents following the monetary exchange strategy. The patterns observed in Figures S8 and S9 closely mirror those in Figure 4 of the main text. In particular, we again observe the excess liquidity effect i.e., a decrease in cooperation due to the proliferation of defectors at very high liquidity levels when the returns to cooperation are low. For the remaining parameter combinations, the monetary exchange strategy remains dominant, with cooperation rates generally increasing with liquidity.

Interestingly, in the absence of the standing-based indirect reciprocity strategy, the upper-right corner of the parameter space, corresponding to combinations where liquidity is very low and the returns to cooperation are very high, no longer features cooperators within a mix of stable equilibrium strategies, but only the monetary exchange strategy. It is worth recalling that under these same conditions in Figure 4 of the main manuscript, we observed a takeover by a mix of indirect reciprocators and unconditional cooperators. Here, however, without indirect reciprocators refusing to cooperate with money users who belong to a sparse monetary circuit due to low liquidity, unconditional cooperators fail to spread.

Liquidity	1.25	1.5	2	3	4	5	10	20	50	100
0.25	0.20 (0.04)	0.22 (0.02)	0.22 (0.02)	0.21 (0.03)	0.22 (0.02)	0.22 (0.02)	0.22 (0.03)	0.22 (0.02)	0.22 (0.02)	0.22 (0.03)
0.5	0.37 (0.03)	0.38 (0.03)	0.37 (0.03)	0.37 (0.03)	0.37 (0.03)	0.37 (0.03)	0.37 (0.03)	0.37 (0.03)	0.38 (0.03)	0.37 (0.03)
1	0.58 (0.03)	0.58 (0.03)	0.57 (0.03)	0.57 (0.02)	0.57 (0.03)	0.58 (0.03)	0.57 (0.03)	0.58 (0.02)	0.58 (0.03)	0.58 (0.03)
2	0.76 (0.02)	0.76 (0.02)	0.75 (0.02)	0.75 (0.03)	0.75 (0.03)	0.75 (0.02)	0.76 (0.03)	0.75 (0.03)	0.76 (0.03)	0.76 (0.02)
3	0.83 (0.02)	0.84 (0.02)	0.83 (0.02)	0.83 (0.02)	0.83 (0.02)	0.83 (0.02)	0.84 (0.02)	0.84 (0.02)	0.84 (0.02)	0.84 (0.02)
5	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)
10	0.95 (0.10)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)
15	0.96 (0.28)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)
20	0.97 (0.34)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.97 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)
25	0.95 (0.47)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)
35	0.00 (0.38)	0.99 (0.25)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)
50	0.00 (0.00)	0.98 (0.49)	0.99 (0.00)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)	0.99 (0.00)	0.99 (0.01)
75	0.00 (0.00)	0.00 (0.25)	0.99 (0.22)	0.99 (0.01)	0.99 (0.00)	0.99 (0.00)	0.99 (0.00)	0.99 (0.00)	0.99 (0.00)	0.99 (0.00)
100	0.00 (0.00)	0.00 (0.00)	0.99 (0.50)	1.00 (0.10)	1.00 (0.00)	1.00 (0.00)	1.00 (0.00)	0.99 (0.00)	1.00 (0.00)	1.00 (0.00)

Benefit-Cost Ratio

Figure S8. Cooperation rates and dominant strategies at the end of the simulation (30000th step) across 140 combinations of benefit-to-cost ratios and liquidity values. Reciprocity strategies are excluded in this scenario. The initial population consists of 167 unconditional cooperators, 167 defectors, and 167 agents following the monetary exchange strategy. Larger numbers indicate the median cooperation rate for each combination of benefit-to-cost ratio and liquidity level, while smaller numbers in parentheses indicate the standard deviations over 100 independent runs. Cell colors represent the dominant strategy: green for money and yellow for defectors. A transparency gradient visually illustrates mean cooperation rates across the parameter space, with darker shades indicating higher cooperation.

Liquidity \ Benefit-Cost Ratio	1.25	1.5	2	3	4	5	10	20	50	100
0.25	0.19 (0.04)	0.22 (0.02)	0.22 (0.02)	0.21 (0.02)	0.21 (0.02)	0.21 (0.02)	0.22 (0.02)	0.21 (0.02)	0.21 (0.02)	0.21 (0.02)
0.5	0.36 (0.03)	0.38 (0.02)	0.37 (0.03)	0.38 (0.02)	0.37 (0.02)	0.37 (0.03)	0.37 (0.02)	0.38 (0.02)	0.38 (0.03)	0.38 (0.02)
1	0.57 (0.03)	0.57 (0.02)	0.57 (0.02)	0.58 (0.03)	0.57 (0.03)	0.58 (0.02)	0.57 (0.02)	0.58 (0.03)	0.57 (0.02)	0.57 (0.02)
2	0.75 (0.02)	0.76 (0.02)	0.76 (0.02)	0.76 (0.02)	0.76 (0.02)	0.76 (0.02)	0.76 (0.02)	0.75 (0.02)	0.76 (0.02)	0.75 (0.02)
3	0.83 (0.02)	0.84 (0.02)	0.83 (0.02)	0.83 (0.02)	0.83 (0.02)	0.83 (0.02)	0.83 (0.02)	0.84 (0.02)	0.83 (0.02)	0.83 (0.02)
5	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)	0.90 (0.01)	0.90 (0.02)
10	0.95 (0.02)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)
15	0.96 (0.13)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)
20	0.97 (0.19)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.98 (0.01)	0.97 (0.01)	0.97 (0.01)	0.98 (0.01)	0.97 (0.01)
25	0.97 (0.35)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)
35	0.01 (0.46)	0.98 (0.10)	0.99 (0.01)	0.98 (0.01)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)	0.98 (0.01)	0.98 (0.01)	0.99 (0.01)
50	0.00 (0.17)	0.99 (0.37)	0.99 (0.01)	0.99 (0.00)	0.99 (0.00)	0.99 (0.01)	0.99 (0.00)	0.99 (0.00)	0.99 (0.01)	0.99 (0.01)
75	0.00 (0.00)	0.00 (0.38)	0.99 (0.10)	0.99 (0.00)	0.99 (0.00)	0.99 (0.00)	0.99 (0.00)	0.99 (0.00)	0.99 (0.00)	0.99 (0.00)
100	0.00 (0.00)	0.00 (0.10)	0.99 (0.38)	0.99 (0.00)	0.99 (0.00)	0.99 (0.00)	0.99 (0.00)	0.99 (0.00)	0.99 (0.00)	0.99 (0.00)

Figure S9. Cooperation rates and dominant strategies at the end of the simulation (30000th step) across 140 combinations of benefit-to-cost ratios and liquidity values. Reciprocity strategies are excluded in this scenario. The initial population consists of 200 unconditional cooperators, 200 defectors, and 200 agents following the monetary exchange strategy. Larger numbers indicate the median cooperation rate for each combination of benefit-to-cost ratio and liquidity level, with smaller numbers in parentheses indicating the standard deviations over 100 independent runs. Cell colors represent the dominant strategy: green for money and yellow for defectors. A transparency gradient visually illustrates mean cooperation rates across the parameter space, with darker shades indicating higher cooperation.

Finally, Figure S10 reproduces Figure 4 of the main manuscript, but now implements an indirect reciprocity strategy based on *image scores* (Nowak & Sigmund, 1998) rather than *standing* (Panchanathan & Boyd, 2003). Under the image-score rule, indirect reciprocators cooperate with randomly matched partners if partners' reputation score, an integer, is positive, and refuse to cooperate if it is zero or negative. Cooperation increases an agent's score by one, while defection decreases it by one. Unlike monetary balances (see the main manuscript), reputation scores are updated for every agent after each action, much like in the standing model, and regardless of the strategy actually used, since reputation operates as a passive, pervasive system of social judgement. At initialization, we assume the image score of every agent is 1.

The key difference from *standing* lies in the failure of *image scoring* to protect the reputation of indirect reciprocators who punish defectors by withholding help. Defectors rapidly accumulate negative scores, which worsen the longer they remain in the simulation, leading indirect reciprocators to refuse to help them. However, this refusal also decreases the reciprocators' own scores. Such downward spirals also occur when indirect reciprocators withhold help from direct reciprocators who retaliate excessively, or from money users who cooperate too little because liquidity is scarce. This dynamic can trigger a cascade of defection among indirect reciprocators, who begin to punish each other as soon as one's score turns negative. This problem does not arise with *standing*, where the evaluation focuses not on actions per se (whether someone defected) but on intentions — specifically, whether a defection is justified.

Despite this difference, Figure S10 shows that the monetary exchange strategy continues to dominate even when indirect reciprocators follow an image-scoring rule. Similarly to Figures S7 and S9, the excess liquidity effect still emerges when the returns to cooperation are low and liquidity is very high. When liquidity is very low and the returns to cooperation are very high (upper-right corner), image-score reciprocators fail to take over the population together with unconditional cooperators, unlike in the case of standing-based reciprocity (Figure 4 of the main manuscript). This outcome can be attributed to the cascade of defection among indirect reciprocators.

Liquidity \ Benefit-Cost Ratio	1.25	1.5	2	3	4	5	10	20	50	100
0.25	0.19 (0.05)	0.21 (0.03)	0.21 (0.02)	0.22 (0.02)	0.22 (0.02)	0.21 (0.03)	0.22 (0.02)	0.22 (0.02)	0.21 (0.22)	0.22 (0.18)
0.5	0.37 (0.04)	0.38 (0.03)	0.38 (0.02)	0.38 (0.03)	0.38 (0.02)	0.37 (0.03)	0.38 (0.03)	0.37 (0.05)	0.37 (0.07)	0.38 (0.12)
1	0.58 (0.03)	0.58 (0.02)	0.58 (0.03)	0.58 (0.03)	0.58 (0.03)	0.58 (0.03)	0.58 (0.03)	0.58 (0.03)	0.58 (0.05)	0.58 (0.04)
2	0.76 (0.03)	0.75 (0.02)	0.76 (0.02)	0.75 (0.02)	0.75 (0.02)	0.76 (0.02)	0.76 (0.02)	0.76 (0.03)	0.76 (0.04)	0.76 (0.02)
3	0.83 (0.02)	0.84 (0.02)	0.83 (0.02)	0.84 (0.02)	0.83 (0.02)	0.83 (0.02)	0.83 (0.02)	0.83 (0.02)	0.83 (0.02)	0.84 (0.02)
5	0.90 (0.08)	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)	0.90 (0.02)
10	0.95 (0.16)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)	0.95 (0.01)
15	0.96 (0.36)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)	0.97 (0.01)
20	0.97 (0.44)	0.98 (0.14)	0.98 (0.01)	0.97 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.97 (0.01)
25	0.80 (0.48)	0.98 (0.19)	0.98 (0.10)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)	0.98 (0.01)
35	0.00 (0.40)	0.99 (0.31)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)
50	0.00 (0.00)	0.98 (0.49)	0.99 (0.17)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)	0.99 (0.01)
75	0.00 (0.00)	0.00 (0.35)	0.99 (0.32)	0.99 (0.00)	0.99 (0.01)	0.99 (0.00)	0.99 (0.01)	0.99 (0.00)	0.99 (0.00)	0.99 (0.00)
100	0.00 (0.00)	0.00 (0.10)	0.99 (0.49)	1.00 (0.10)	0.99 (0.00)	1.00 (0.01)	1.00 (0.00)	0.99 (0.00)	0.99 (0.00)	1.00 (0.00)

Figure S10. Cooperation rates and dominant strategies at the end of the simulation (30000th step) across 140 combinations of benefit-to-cost ratios and liquidity values. Indirect reciprocity is implemented through an image-scoring mechanism. Larger numbers indicate the median cooperation rate for each combination of benefit-to-cost ratio and liquidity level, with smaller numbers in parentheses indicating the standard deviations over 100 independent runs. Cell colors represent the dominant strategy: green for money and yellow for defectors. A transparency gradient visually illustrates mean cooperation rates across the parameter space, with darker shades indicating higher cooperation.

References

Nowak, M. A., & Sigmund, K. (1998). Evolution of indirect reciprocity by image scoring. *Nature*, *393*(6685), 573-577. https://doi.org/10.1038/31225

Panchanathan, K., & Boyd, R. (2003). A tale of two defectors: the importance of standing for evolution of indirect reciprocity. *Journal of Theoretical Biology*, *224*(1), 115-126. https://doi.org/10.1016/S0022-5193(03)00154-1.